c) The dimension of Nul A is the number of variables in the equation $Ax=0$.

Soln: By Example 2, we know that the dimension of Nul A is the number of free variables in the equation $Ax=0$. ∴ The given statement is False.

d) The dimension of the column space of A is rank A.

Soln: This is true by the defn. given on Page 157.

e) If H is a p-dimensional subspace of $\mathbb{R}^n$, then a linearly independent set of p vectors in H is a basis for H.

Soln: By Theorem 15 (Basis Theorem) on Page 158, we know that the given statement is True. □

In Exercises 19-24, justify each answer or construction.

19) If the subspace of all solutions of $Ax=0$ has a basis consisting of three vectors and if A is a 5×7 matrix, what is the rank of A?

Soln: By defn, the subspace of all solutions of $Ax=0$ is called the nullspace of A. The nullspace of A has a basis of 3 vectors. ∴ From Example 2, the number of free variables in $Ax=0$ is 3. ∴ The no. of pivot cols is $7-3=4$. By Theorem 13, $\dim \text{Col}\, A = 4$. By defn, $\text{rank}\, A = 4$.

20) What is the rank of a 4×5 matrix whose null space is three-dimensional?

Soln: By a solution similar to Prob 19, rank = 2.

21) If the rank of a 7×6 matrix A is 4, what is the dimension of the solution space of $Ax=0$?

Soln: By defn, $\dim \text{Col}\, A = 4$. By defn, dim Col A represents the cardinality of the basis of Col A.

By Theorem 13, the pivot columns of A represents the basis of Col A.
∴ The no. of pivot columns of A is 4. So, the no. of free variables of $Ax=0$ is $6-4=2$.

By defn, Nullspace of A represents the solution space of $Ax=0$.

By Theorem 12, Nul A is a subspace of $\mathbb{R}^6$.

By defn. of dimension and Example 2, dimension of the solution space of $Ax=0$ is 2. □